Add-On Data and Investment Decisions

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Abstract

Does paying for market data change investor behavior?

Keywords: experimental finance, market data, retail trading, order imbalance

JEL Codes: G11, G12, G14

1 Introduction

> "No matter what sort of data you're after — Forex, money market, fixed income, or equity — you need to acquire it. Even if it is a one-off thing for your trade, you need to acquire it."
>
> — Interactive Brokers Campus, "What Every Trader Should Know About Market Data"

Does paying for market data change how individual investors interpret and use it? Retail brokerages such as Robinhood, Interactive Brokers, or TD Direct Investing increasingly un-bundle market data from their core trading services, selling near-real-time order flow and market depth as an add-on for prices ranging between \$5 and \$50 per month.[1] Order flow is valuable for hedge funds with the speed and infrastructure to extract short-term signals (Bartlett et al., 2023; Kwan et al., 2024), but retail traders lack these capabilities. Having paid for data, traders may overtrade on it to justify the cost: The act of paying may itself alter how traders value the information.

The question matters because retail investors now account for a substantial, and rapidly growing, share of market activity. In April 2025, individual traders represented 36% of U.S. equity and options order flow, up from under 10% before the pandemic.[2] The share is higher still in short-term derivatives: retail accounts for roughly 60% of same-day (0DTE) options volume (Bryzgalova et al., 2023), now more than half of all S&P 500 options activity. As retail participation grows, so does the importance of understanding how these traders respond to platform design — including add-on data products sold separately from core brokerage services.

We conjecture that the act of *investment* in market data shapes the beliefs and strategies of retail traders. Specifically, we hypothesize an endowment effect driven by endogenous information acquisition. By purchasing access to market feeds, investors come to treat the data as an asset they own rather than as a neutral input. This sense of ownership increases the perceived signal value of complex market data, leading investors to rely on it more heavily when forming beliefs. As a result, beliefs respond more strongly to noisy data and pass through more aggressively into portfolio choices — even when the data itself has no predictive value given investors' available tools.

These effects should be heterogeneous across investors, generating selection into data acquisition. Overconfident investors, in particular, should be more likely both to purchase data ex ante and to overestimate

[1] See Chapkovski et al. (2025b) for a detailed snapshot of the retail market data landscape.

[2] See J.P. Morgan: Who is buying U.S. equities (June 4, 2025); Financial Times: Are retail traders the captains now? (December 9, 2025)

their ability to extract value from it ex post, amplifying belief distortions and the pass-through from beliefs to portfolio decisions.

This paper provides the first controlled evidence on how actively paying for market data affects belief formation and investment behavior. We build on recent work by Andries et al. (2025) and develop an online randomized controlled experiment in oTree (Chen et al., 2016), in which participants complete a sequence of independent investment rounds. In each round, participants observe a graphical display of 16 historical realizations of stock returns and order-flow imbalance, mapped to half-hour intervals within a trading day. Using this information, participants form beliefs about next-period returns and choose how much of their endowment to invest in the risky asset.

Crucially, the informational content of order flow varies across rounds. In predictable rounds, lagged order imbalance forecasts next-period returns with a correlation of approximately 0.75; in baseline rounds, the imbalance is pure noise and carries no predictive content. Participants are informed that the data may or may not be predictive but are never told which rounds are predictable, forcing them to infer signal quality from the data itself.

To isolate the causal effect of paying for information from selection effects, we introduce a two-stage design. Treatment participants choose each round between paying E$5 for accessing order-flow data or receiving no data. After they choose, we randomly draw the round type: either the choice are binding (paid data condition), or all participants receive data for free regardless of their choice (free data condition). Control participants always receive free data without being asked. This design generates random variation in payment among participants who chose data, cleanly identifying the sunk cost effect. The free data condition also allows us to observe beliefs for participants who declined to pay, separating selection from treatment effects.

2 Literature review

We contribute to the experimental literature on individual investors in modern financial markets. Closest to our paper, Andries et al. (2025) show that when predictive signals are transparent and freely available, investors update forecasts in the right direction but underreact in their portfolio choices. At the same time, Chapkovski et al. (2025b) document that retail investors are willing to pay more for data than the value they can extract from it. We instead isolate the behavioral consequences of actively paying for information. In our design, participants in the costly and free data conditions observe identical order-flow histories; the only difference is whether access requires an active purchase decision. This allows us to cleanly identify how *investment* in data, rather than the data itself, shapes beliefs and investment behavior.

Beyond identifying the causal effect of paying for information, our design also sheds light on who pays for data, contributing to the literature on endogenous information acquisition (Huber et al., 2011; Fuster et al., 2022). By observing voluntary purchase decisions, we characterize selection into data acquisition and show how overconfidence, financial literacy, and risk preferences shape willingness to pay for complex market data.

Our paper is also related to work studying how platform architecture influences retail trading behavior. Chapkovski et al. (2026, 2025a) show that gamification increases trading volume and risk-taking and can reinforce trading mistakes, while Yelagin (2024) finds that gamification raises return volatility but makes retail order flow less toxic. Other studies document that engagement tools such as price notifications, alerts, and smartphone interfaces increase attention and risk-taking (Arnold et al., 2022; Kalda et al., 2021), and that attention-steering features on Robinhood contribute to poor performance (Barber et al., 2022). We complement this literature by studying a distinct and increasingly important platform feature: the sale of granular market data to retail traders.

We further contribute to the literature on retail trading and investor heterogeneity. Earlier studies show that retail trades can predict returns, with aggressive trades anticipating news and passive orders providing contrarian liquidity (Kaniel et al., 2008; Kelley and Tetlock, 2013). More recent evidence highlights substantial heterogeneity across retail investors, with order flow on platforms such as Robinhood differing in sophistication and predictive content from broader retail flow (Eaton et al., 2022). Our design speaks directly to this heterogeneity by measuring both selection into data acquisition and belief updating conditional on acquisition, using financial literacy and overconfidence as proxies for investor sophistication.

Third, we contribute to the literature on the value of financial data and endogenous information acquisition. Recent work treats data as an economic asset whose value depends on how it improves decision-making and equilibrium outcomes (Veldkamp, 2023; Farboodi et al., 2024). By holding the information environment fixed and varying only whether investors actively pay for access, our experiment allows us to compare willingness to pay with the value investors can extract from identical data. We show that paying for information can distort belief formation and investment responses, generating systematic overvaluation even when processing capacity is unchanged.

Finally, our hypotheses relate to the endowment effect, which shows that ownership can distort valuation even when it does not affect underlying payoffs. Classic experiments document that individuals assign higher value to goods they own, creating gaps between willingness to pay and willingness to accept (Kahneman et al., 1990), with effects strongest among inexperienced agents and attenuating with experience (List, 2011). While this literature focuses on physical or financial assets, we study an endowment effect in the use of information. In our setting, paying for data creates a sense of ownership over an informational input, increasing its perceived precision and its influence on beliefs and portfolio choices. Our design directly tests this mechanism by holding information fixed and varying only whether access requires an active purchase decision.

3 Experiment design

3.1 Round investment game

Our experimental design builds on the framework of Andries et al. (2025). In each round, participants observe past realizations of two variables: (i) a "stock return" r_t and (ii) an order flow imbalance ω_t. The display consists of 16 past observations — visually mapped to half-hour intervals over a trading day — to mimic an intraday trading environment.

Across rounds, the order imbalance may or may not predict returns. Participants know this but are not told which specific rounds contain predictive information, nor the fraction of such rounds (set to 50%). We refer to rounds where the imbalance is uninformative as *baseline* rounds, and to those where it predicts returns as *predictable* rounds.

In baseline rounds, stock returns are drawn from a random walk process

$$r^b_{t+1} = \mu + \epsilon^b_{t+1}, \tag{1}$$

where $\mu = 15\%$ and $\epsilon^b_t \sim \mathcal{N}\left(0, \sigma^2_b\right)$ with $\sigma = 12\%$. Conversely, in predictable rounds, stock returns load on the lagged order imbalance:

$$r^p_{t+1} = \mu + \omega_t + \epsilon^p_{t+1}, \tag{2}$$

where the order imbalance is identically and independently distributed as $\omega_t \sim \mathcal{N}\left(0, \sigma_\omega^2\right)$ and $\epsilon_t^p \sim \mathcal{N}\left(0, \sigma_p^2\right)$. We impose $\sigma_p^2 + \sigma_\omega^2 = \sigma_b^2$ to ensure that r^b and r^p share the same unconditional distribution. The only difference between baseline and predictable rounds is thus the presence of an informative signal that is correlated with returns.

In the experiment, we set $\sigma_\omega = 9\%$ which implies $\sigma_p = 7.9\%$. This parametrization yields a correlation of $\text{Corr}\left(r_{t+1}^p, \omega_t\right) = \frac{0.09^2}{0.12 \times 0.09} = 0.75$, ensuring that the imbalance is sufficiently informative to be visually detectable in the time-series display, while still leaving substantial residual noise. This mirrors realistic intraday settings, where order-flow imbalance carries predictive content but is far from a perfect signal. Figure 1 illustrates the information displayed to participants in each round.

At the start of each round, participants receive an endowment of 100 E\$ (experimental dollars). They choose how much to invest in the risky stock; the remainder is held in cash at a zero return. Leverage and short selling are not permitted. In addition to choosing their portfolio, participants report (i) whether they believe the order-flow imbalance is informative in that round and (ii) their forecast of the next-period return.

At the end of the round, participants receive feedback on the realized return and on whether the imbalance was in fact informative. They earn E\$2 for correctly identifying the round type and an additional E\$2 for providing a forecast within one percentage point of the realized return.

3.2 Estimating the value of order imbalance data

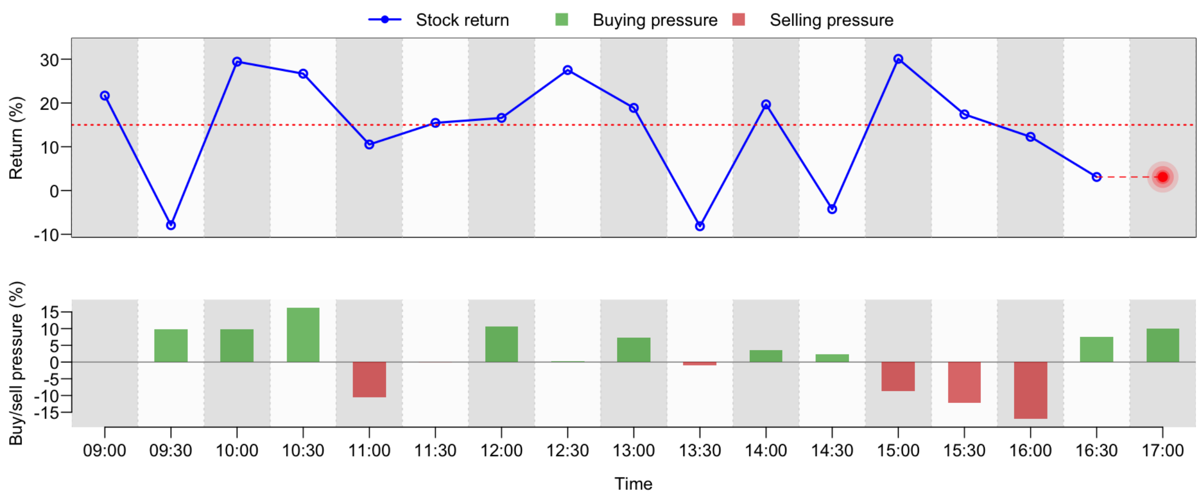
To quantify the ex ante value of data, we impose structure on investor preferences. We assume investors have constant relative risk aversion (CRRA) over wealth, $U(w) = \frac{w^{1-\gamma}}{1-\gamma}$, where γ is the risk aversion coefficient.

Under CRRA preferences and normally distributed returns, the classical result in Merton (1969) implies that investors allocate a constant share of wealth to the risky asset equal to $\alpha = \frac{\mathbb{E}r}{\gamma\sigma^2(r)}$. Using our experimental parameters and a risk-aversion level of $\gamma = 24$ (calibrated to estimates in Andries et al., 2025), the optimal portfolio share in baseline rounds is $\alpha_b^\star = 43.4\%$. This allocation yields a certainty equivalent of E\$103.31 for an initial endowment of E\$100.

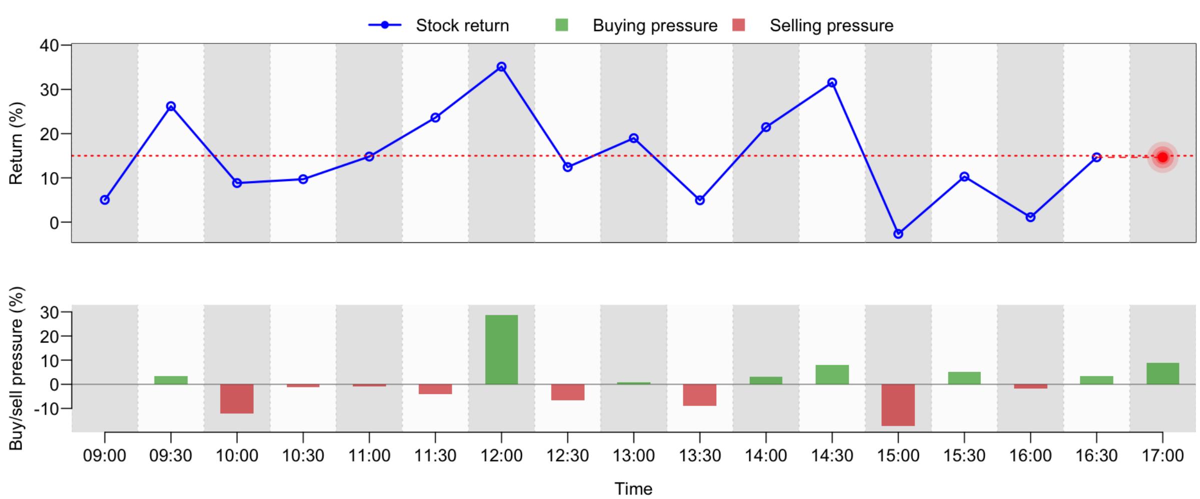
To estimate optimal investment in predictable rounds, we simulate 20,000 realizations of the order-flow

Figure 1: **Example screens shown to participants.**

Each panel displays 16 past half-hour returns and order-flow imbalances (buy/sell pressure). The return to be predicted (at 17:00) is marked by a red fuzzy circle.



(a) Baseline round: order imbalance is uninformative.



(b) Predictable round: order imbalance is informative.

imbalance. For each simulated draw ω_i, the optimal risky share is

$$\alpha^{\star}_{r,i} = \max\left\{0, \min\left\{\frac{\mu + \omega_i}{\gamma\sigma_p^2}, 1\right\}\right\}, \tag{3}$$

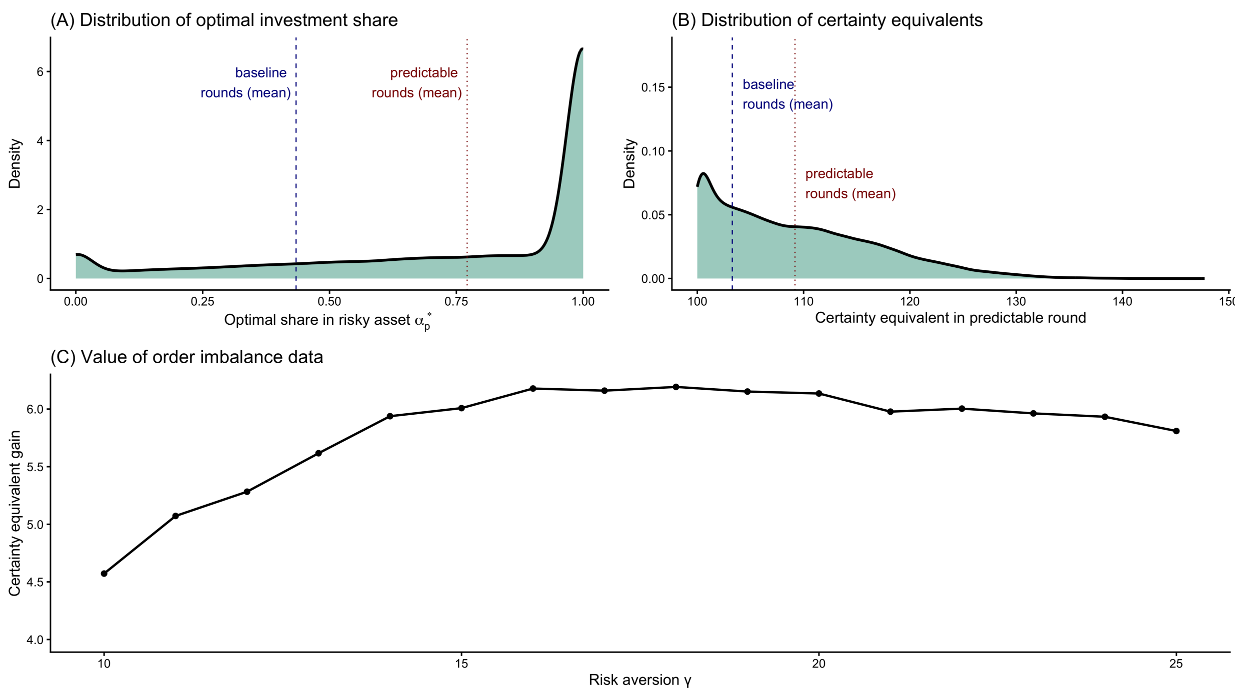
that is, the Merton share projected onto the admissible interval $[0, 1]$ reflecting the no-leverage and no-shorting constraints.

Figure 2 illustrates the simulation results for $\gamma = 24$. Panel (A) plots the distribution of optimal risky shares $\alpha^{\star}_{p,i}$, which vary substantially across simulations. The mean optimal share is 77%: that is, an increase of roughly 62 percentage points relative to the baseline allocation. Predictive information therefore leads investors to take substantially larger positions on average, and it also generates considerable dispersion.

Panel (B) reports the corresponding distribution of certainty equivalents (CEs) across the $20,000$ predictable-round simulations. The distribution is right-skewed: while most realizations yield moderate improvements over the baseline CE of E\$103.31, a meaningful subset delivers much larger utility gains when the order-flow signal strongly increases expected returns. The mean and median certainty equivalent in predictable rounds are E\$109.18, implying an ex-ante value of data of E\$5.87 (=E\$109.18-E\$103.31).

Figure 2: **Optimal portfolios and the value of predictive order flow data**

This figure shows simulation results for the optimal risky share and certainty-equivalent gains under CRRA preferences. Panel (A) plots the distribution of optimal risky shares in predictable rounds. Panel (B) reports the corresponding distribution of certainty equivalents. Panel (C) shows the ex ante value of data as a function of risk aversion γ.



Panel (C) summarizes the ex ante value of order-flow imbalance data as a function of risk aversion γ. The certainty-equivalent gain is strictly positive for all γ considered, confirming that predictive order-flow information is always valuable, and it remains relatively stable between E$5 and E$6. On the one hand, data is valuable for risk-averse investors because it reduces the residual noise in future returns. On the other hand, as risk aversion increases, optimal positions shrink both with and without data, which dampens the incremental benefit of information.

3.3 Experiment design

Each participant plays 14 rounds, starting each round with an endowment of E$100. The first two rounds are training rounds in which all participants observe order-flow data.

We randomly assign participants to a **treatment** or **control** group. Control participants observe order-flow data at no cost in all twelve non-training rounds.

In the treatment group, participants may purchase order-imbalance data for E$5 (the calibrated fair value from Section 3.2) at the start of each round — following, e.g., Huber et al. (2011); Fuster et al. (2022). After they make a choice, they are randomly assigned to one of two possible conditions:

- **Paid data round** (drawn with probability $\pi = 2/3$). In paid data rounds, the purchase decision is binding. Participants who chose to pay for data see the order-flow imbalance and have their balance reduced by E$5. Those who decline see the message: "*You do not have privileges to access order-flow data.*"
- **Free data** (probability $1/3$). In free data rounds, all participants see order-flow data at no cost regardless of their choice.

Order-flow data is informative in exactly half of the rounds; however, participants are not told the share of informative rounds.

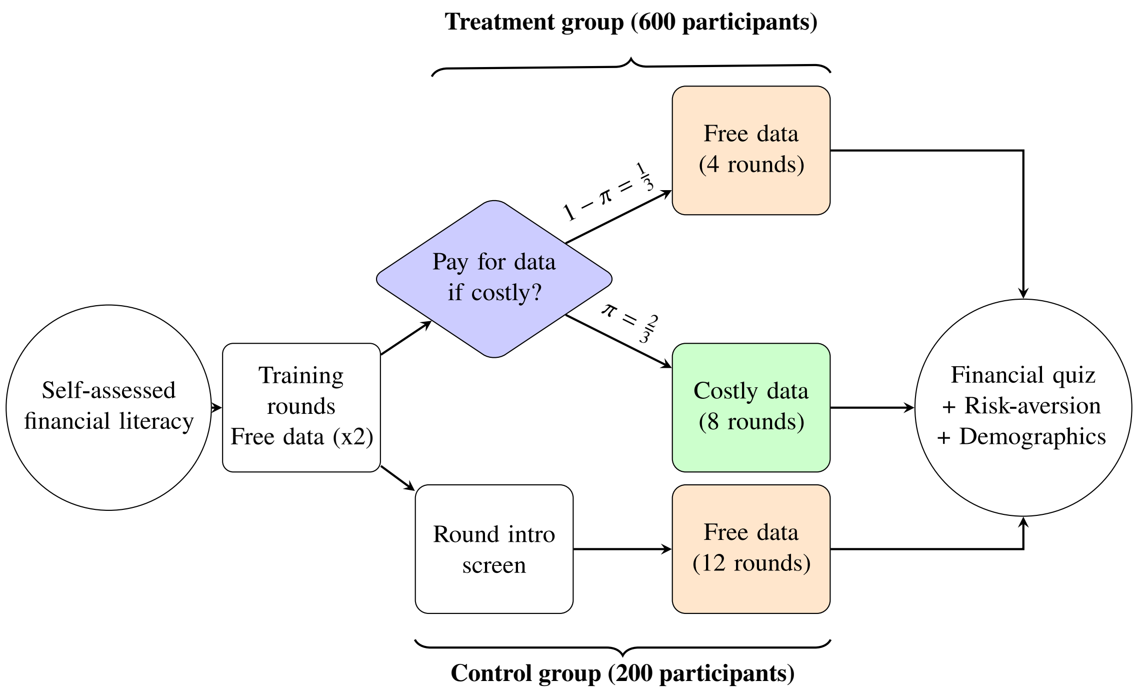
Finally, participants answer 13 financial literacy questions provided in Appendix A. The questions are adapted from Fernandes et al. (2014) and include the three-question measure developed by Lusardi and Mitchell (2011), as well as two trading specific questions following the MiFID questionnaire in Inghelbrecht and Tedde (2024). To measure overconfidence and distinguish between subjective and objective financial literacy, we also ask participants:

> *On a scale from zero to ten, where zero is not at all knowledgeable about personal finance and ten is very knowledgeable about personal finance, what number would you be on the scale?*

This question measures self-assessed (subjective) financial knowledge and is identical to the one used in Cupák et al. (2020). We pose this question before the experiment begins to avoid any influence from the trading game's monetary performance or the quiz's perceived difficulty.

Figure 3 illustrates the experiment's timeline. Appendix B reproduces the consent form. The experimental instructions given to participants are reproduced in Appendix C. Before the trading starts, participants need to correctly answer six comprehension questions, listed in Appendix D. This allows us to make sure that participants indeed understand the experiment before the trading rounds start. Finally, at the end of the experiment all participants are required to fill in a risk-aversion (Holt and Laury, 2002) and a demographic questionnaire.

Figure 3: **Experiment timing**



4 Hypotheses development

Our framework isolates three distinct forces shaping beliefs about order-flow data:

(i) *Sunk cost effect.* Paying for data distorts beliefs and decisions *ex post*, holding the information set fixed.

(ii) *Selection effect.* Participants differ in their *ex ante* willingness to pay depending on financial literacy, overconfidence, and risk aversion.

(iii) *Elicitation effect.* Being asked whether to pay may itself affect beliefs, even for participants who decline.

Since participants in treatment and control groups receive identical price and order flow information, differences in beliefs or investment behavior must arise from the act of purchasing the data rather than from differential information sets.

Hypothesis 1. ***Payment and beliefs.*** *Participants who pay for order-flow data are more likely to perceive order-flow imbalance as informative, particularly when imbalance is objectively uninformative.*

The intuition is straightforward. Once participants pay the data fee, they become motivated to believe the expenditure was worthwhile. The purchased signal is mentally coded as an "asset" they now own, which inflates its perceived value and predictive content. As a result, participants who pay are more likely to interpret noise as information and to assign higher subjective informativeness to identical order-flow patterns. To test Hypothesis 1, we estimate:

$$\begin{aligned} B_{it} = \alpha &+ \beta_1 \text{Treated}_i + \beta_2(\text{PaidRound}_t \times \text{Informative}_t) + \beta_3(\text{PaidRound}_t \times (1 - \text{Informative}_t)) \\ &+ \beta_4 \text{Informative}_t + \beta_5 \text{PayChoice}_{it} + X_i'\gamma + \phi \text{Round}_t + \varepsilon_{it}, \end{aligned} \tag{4}$$

where B_{it} equals one if participant i perceives order flow as informative in round t; *Treated*$_i$ indicates treatment assignment; *PayChoice*$_{it}$ indicates choosing to pay for data; *PaidRound*$_{it}$ indicates a paid data round, and *Informative*$_t$ indicates whether order flow truly predicts returns. All specifications control for round number and participant demographics X_i.

The coefficient β_2 and β_3 identify the effect. Identification relies on random assignment: among participants with *PayChoice*$_{it} = 1$, whether *PaidRound*$_{it} = 1$ or *PaidRound*$_{it} = 0$ is determined by a lottery draw. Hypothesis 1 predicts $\beta_3 > 0$. We also expect $\beta_2 > 0$ although the effect is likely to be weaker, as participants may more readily recognize informativeness when order flow is genuinely predictive.

Hypothesis 2. ***Payment and forecasts.*** *Conditional on perceiving imbalance as informative, participants who pay for order-flow data adjust their return forecasts more strongly to order-flow imbalance.*

If payment inflates the perceived value of order-flow information, it should not only affect whether participants view imbalance as informative, but also how strongly they incorporate it into their stated forecasts. Conditional on perceiving imbalance as informative, participants who pay for data may place greater weight on order-flow imbalance when forming return forecasts. To test Hypothesis 2, we estimate:

$$\begin{aligned} F_{it} = {} & \alpha + \delta_1\left(\text{PaidRound}_t \times B_{it} \times \text{Imbalance}_t\right) + \delta_2\left(\text{PaidRound}_t \times (1 - B_{it}) \times \text{Imbalance}_t\right) \\ & + \delta_3\left(B_{it} \times \text{Imbalance}_t\right) + \delta_4 B_{it} + \delta_5\left(\text{PaidRound}_t \times B_{it}\right) + \delta_6\left(\text{PaidRound}_t \times (1 - B_{it})\right) \\ & + \delta_7\,\text{PayChoice}_{it} + \delta_8\left(\text{PayChoice}_{it} \times \text{Imbalance}_t\right) + \delta_9\,\text{Treated}_i + \delta_{10} r_t + \delta_{11}\text{Imbalance}_t \\ & + \phi\,\text{Round}_t + X_i'\zeta + \nu_{it}. \end{aligned} \tag{5}$$

Hypothesis 2 predicts $\delta_1 > 0$: participants who pay for data adjust their forecasts more strongly to order-flow imbalance when they perceive the signal as informative.

Hypothesis 3. ***Sunk-cost investment.*** *Participants who pay for order-flow data invest larger amounts than participants who receive identical data for free.*

Once participants incur a non-recoverable data fee, they face pressure to justify the expenditure through action. While information can in principle improve decision quality, sunk-cost bias operates through commitment to visible behavior rather than optimal updating (Staw, 1976; Arkes and Blumer, 1985). In our setting, the most tangible way to "use" purchased data is to trade on it, which provides an immediate sense of having extracted value from the purchase.

We formalize this as in Prelec and Loewenstein (1998), by introducing the psychological burden of payment ("pain of paying") into the investor utility function. Investors choose investment level θ to maximize

$$\max_{\theta} \mathbb{E}\left[U\left(W + \theta r\right) \mid m\right] - \lambda \ell\left(\theta\right), \tag{6}$$

where m denotes the signal used in forecasting returns, $\lambda \in \{0, 1\}$ indicates whether the data cost is incurred, and $\ell(\theta)$ is the sunk cost function capturing the disutility from under-utilizing the purchased data. We assume that investing more reduces the sense of waste ($\ell'(\theta) < 0$) and additional investments generate diminishing marginal relief ($\ell''(\theta) > 0$). The first-order condition is:

$$f(\lambda, \theta) = \mathbb{E}\left[U'(W + \theta r) r \mid s\right] - \lambda \ell'(\theta) = 0. \tag{7}$$

Applying the implicit function theorem:

$$\frac{\partial \theta^{\star}}{\partial \lambda} = -\frac{\ell'(\theta)}{\mathbb{E}\left[U''(W + \theta r) r^2 \mid s\right] - \lambda \ell''(\theta)} > 0. \tag{8}$$

since the numerator is positive ($-\ell' > 0$) and the denominator is negative (second-order condition). Thus, sunk-cost pressure increases equilibrium investment to justify the purchase of data.

Hypothesis 4. ***Belief pass-through.*** *Participants who pay for order-flow data exhibit lower sensitivity of portfolio choices to their own return forecasts than participants who receive identical data for free.*

The sunk-cost mechanism not only raises average investment but also reduces the elasticity of investment to beliefs. The rationale is that the psychological pull toward "using" the data increases investment levels but at the same time dampens how much beliefs and forecasts translate into position changes.

To see this formally, note that the first-order condition implicitly defines optimal investment $\theta^{\star}$ as a function of the forecast m and sunk-cost indicator λ. Applying the implicit function theorem, the sensitivity of investment to beliefs is:

$$\left|\frac{\partial \theta^{\star}}{\partial m}\right| = \frac{\left|f_m\right|}{\lambda \ell''(\theta) - \mathbb{E}\left[U''(W + \theta r) r^2 \mid s\right]}, \tag{9}$$

where f_m captures how the marginal value of investment responds to beliefs. The denominator is positive by the second-order condition. For investors who incurred the data cost ($\lambda = 1$), the additional term $\ell''(\theta) > 0$ increases the denominator, strictly reducing belief sensitivity. The curvature of the sunk-cost term acts like additional risk aversion specific to deviating from a "justifying" investment level.

To test Hypotheses 3 and 4, we estimate the sensitivity of investment to return forecasts separately for paid and free-data participants. We use an instrumental variable approach to extract "informed forecasts" as in Andries et al. (2025). Specifically, we instrument stated return forecasts with the order-flow imbalance and last stock return displayed during the experiment. We estimate:

$$\theta_{it} = \beta_1 \widehat{F}_{it} + \beta_2 (\widehat{F}_{it} \times \text{PaidRound}_t) + \gamma \text{PaidRound}_t + \mathbf{X}_{it}' \delta + \varepsilon_{it}, \tag{10}$$

where θ_{it} is participant i's investment in round t, $\widehat{F}_{it}$ is the instrumented return forecast, $PaidRound_t$ indicates whether the participant paid for data access in round t, and $\mathbf{X}_{it}$ includes round fixed effects and additional controls. We restrict the sample to the informative-signal condition, where order flow genuinely predicts returns and belief sensitivity is economically meaningful. The coefficient β_1 captures belief pass-through for free-data participants, while β_2 tests whether paid participants exhibit differential sensitivity. Hypothesis 3 predicts $\gamma > 0$: paying for data increases average investment as participants seek to justify the expenditure. Hypothesis 4 predicts $\beta_2 < 0$: paying for data attenuates the responsiveness of investment to beliefs.

An alternative mechanism is that payment for data induces participants to rely more on it when making investment decisions. If this is the case, we should observe no shift in investment levels upon data payment and a *higher* sensitivity of investment to forecasts: in equation (10), we would expect $\beta_2 > 0$ and $\gamma = 0$.

Hypothesis 5. ***Selection effects.*** *Overconfident participants, risk-averse participants, and participants with higher financial literacy are more likely to purchase order-flow data.*

The decision to purchase data reflects heterogeneous valuations of information. Overconfident individuals may overestimate their ability to translate information into profitable trades. Financially literate participants may value data for rational reasons, recognizing how order-flow signals can improve forecasts. To test Hypothesis 5, we estimate:

$$\text{PayChoice}_{it} = \alpha + \beta_1 \text{Overconfidence}_i + \beta_2 \text{FinQuiz}_i + \beta_3 \text{RiskAversion}_i + \phi \, \text{Round}_t + X_i' \zeta + \varepsilon_{it}, \tag{11}$$

Hypothesis 5 predicts $\beta_1 > 0$ (overconfidence increases data demand) and $\beta_2 > 0$ (financial literacy increases data demand).

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Online Appendix

January 21, 2026

A Financial literacy quiz

1. Suppose you had $100 in a savings account and the interest rate was 3% per year. After 4 years, how much do you think you would have in the account if you left the money to grow?
   (a) More than $112
   (b) Exactly $112
   (c) Less than $112
   (d) Don't know / Not sure
   (e) Prefer not to say

2. Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, with the money in this account, would you be able to buy
   (a) More than today
   (b) Exactly the same as today
   (c) Less than today
   (d) Don't know / Not sure
   (e) Prefer not to say

3. When placing an order, you may see several prices: the last traded price, the ask price, the bid price, and the closing price. What is the bid price?
   (a) The highest price a buyer is currently willing to pay.
   (b) The price you personally choose when submitting a buy order.
   (c) The lowest price a seller is currently willing to accept.
   (d) The most recent price at which the security traded.
   (e) Don't know / Not sure
   (f) Prefer not to say

4. A 15-year mortgage typically requires higher monthly payments than a 30-year mortgage, but the total interest paid over the life of the loan will be less.
   (a) True
   (b) False
   (c) Don't know / Not sure
   (d) Prefer not to say

5. Normally, which asset described below displays the highest fluctuations over time?
   (a) Savings accounts
   (b) Stocks
   (c) Bonds
   (d) Don't know / Not sure
   (e) Prefer not to say
6. When an investor spreads his money among different assets, does the risk of losing a lot of money:
   (a) Increase
   (b) Decrease
   (c) Stay the same
   (d) Don't know / Not sure
   (e) Prefer not to say
7. Considering a long time period (for example, 10 or 20 years), which asset described below normally gives the highest return?
   (a) Savings accounts
   (b) Stocks
   (c) Bonds
   (d) Don't know / Not sure
   (e) Prefer not to say
8. Do you think that the following statement is true or false? "If you were to invest $1,000 in a stock mutual fund, it would be possible to have less than $1,000 when you withdraw your money."
   (a) True
   (b) False
   (c) Don't know / Not sure
   (d) Prefer not to say
9. When viewing a security's quote, you may see both a bid price and an ask price. What is the bid–ask spread?
   (a) The difference between the lowest ask price and the highest bid price.
   (b) The average of the bid price and the ask price.

(c) The fee the broker charges each time you trade.

(d) The change in the stock price from market open to market close.

(e) Don't know / Not sure

(f) Prefer not to say

10. Which of the following statements is correct?

    (a) Once one invests in a mutual fund, one cannot withdraw the money in the first year

    (b) Mutual funds can invest in several assets, for example invest in both stocks and bonds

    (c) Mutual funds pay a guaranteed rate of return which depends on their past performance

    (d) None of the above

    (e) Don't know / Not sure

    (f) Prefer not to say

11. Which of the following statements is correct? If somebody buys a bond of firm B:

    (a) She owns a part of firm B

    (b) She has lent money to firm B

    (c) She is liable for the company's debts

    (d) None of the above

    (e) Don't know / Not sure

    (f) Prefer not to say

12. Suppose you owe $3,000 on your credit card. You pay a minimum payment of $30 each month. At an annual percentage rate of 12% (or 1% per month), how many years would it take to eliminate your credit card debt if you made no additional new charges?

    (a) Less than 5 years

    (b) Between 5 and 10 years

    (c) Between 10 and 15 years

    (d) Never

    (e) None of the above

    (f) Don't know / Not sure

    (g) Prefer not to say

B Consent form

Dear Participant,

You are invited to take part in a research study on financial decision-making funded by the Social Sciences and Humanities Research Council of Canada. The purpose of this study is to understand how people use data when making investment choices.

Your participation in this study is entirely voluntary. Participation presents no known risks or burdens. You will receive a compensation for participating in the study, announced on the Prolific platform, as well as a monetary compensation proportional to your performance in the experiment. You can withdraw from the study at any time without penalty. However, if you withdraw before the end of the experiment the compensation will be forfeit, as per Prolific's rules.

Please be assured that all data generated during this study will remain confidential. You will be asked to provide some basic demographic information about yourself, but none of this information will in any way be used to personally identify you and no personal information will appear in any published study.

Your decision to participate in this study will remain completely confidential and only the researchers will have access to any of the individual data. We do not collect any data that can personally identify you. Any demographics data will be stored electronically on a web-based server until it is downloaded for the purpose of analysis. This server will only be accessible by the research team using a secure password. Data will be retained in de-identifiable format for 7 years and discarded afterwards. At that time, electronic files will be deleted and written files will be shredded. This is a standard practice for the American Psychological Association. As required by the guidelines of American Economic Association, during the publication process the data might be shared with other investigators for re-analysis. However, to ensure security and confidentiality of the data files only details of the computations sufficient to permit replication might be shared.

This study has been reviewed by the Research Ethics Boards at the University of Calgary. If you are interested, you can receive a summary of the research study by emailing the researchers.

The research study you are participating in may be reviewed for quality assurance to make sure that the required laws and guidelines are followed. If chosen, (a) representative(s) of the Human Research Ethics Program (HREP) may access study-related data and/or consent materials as part of the review. All information accessed by the HREP will be upheld to the same level of confidentiality that has been stated by the research team.

If you wish to obtain further information regarding your rights as a participant you may contact the Conjoint Faculties Research Ethics Board (CFREB) at University of Calgary at cfreb@ucalgary.ca or (403) 220-8640. You may also contact the researchers for further information or to obtain answers to any questions about this research.

I agree to participate in research being conducted by the researchers.

I understand that I can withdraw my consent to participate at any time and by giving consent I am not giving up any of my legal rights. I have read and understood the information above.

C Instructions

Welcome to this trading experiment! Please read these instructions carefully.

Overview. In each round, you will see a graph showing past data for a traded stock, measured over half-hour intervals within a single trading day. The graph displays:

1. **The stock return**, expressed in percentage terms; and
2. **The buy/sell imbalance**, a measure of order flow that captures buying versus selling pressure, expressed in percentage terms. It is positive when buy order flow exceeds sell order flow, and negative when sell order flow exceeds buy order flow.

In some rounds, the buy/sell imbalance is informative and helps predict the next stock return. In other rounds, the two variables are independent, and the imbalance does *not* help predict returns.

Your task. In each round, you are endowed with 100 experimental dollars (E$). Your tasks are:

1. [Treatment only] At the start of each round, you choose whether to purchase access to the buy/sell imbalance data for E$4. After you make your choice, the computer randomly determines the round type:
   - **Paid data round** (probability 2/3): Your choice is binding. If you chose to purchase data, E$4 is deducted from your cash account and you see the imbalance data. If you declined, you pay nothing but do not see the imbalance data.
   - **Free data round** (probability 1/3): You see the imbalance data at no cost, regardless of your initial choice.
2. [Treatment Only] If you see order imbalance data, assess whether the imbalance can be used to predict the next stock return.
3. Provide a forecast for the next stock return.
4. Choose how much of your endowment you would like to invest in the stock.

There are twelve main rounds in the experiment. All rounds are independent, and the average stock return is 15%. Before the main rounds begin, you will complete two practice rounds to familiarize yourself with the interface.

After each round, you will see the realized stock return, whether the buy/sell order flow imbalance was actually informative, the accuracy of your forecast, and the total wealth you earned in that round.

Your payoff. Your final payoff comprises four parts:

1. The value of your portfolio (stocks plus cash) plus any bonus earned, from one randomly selected round.
2. [Treatment only] E$1 for every correct assessment of whether the order-flow imbalance can predict returns.
3. E$1 for every precise forecast (within $\pm 1\%$ of the actual stock return).
4. E$0.25 for each correct answer in a financial literacy quiz which you will complete after the main rounds.

At the end of the experiment, one of the twelve main rounds will be randomly selected to determine your investment payoff (component 1). Your total payoff in ECU is the sum of (1) from the selected round, plus [Treatment only] (2) and (3) accumulated across all rounds, plus (4) from the financial literacy quiz. Your final payment in EUR is your payoff in ECU divided by 20.

D Comprehension quiz

1. (Treatment only) If you choose to purchase data and the round is a **paid data round**, what happens?
   (a) You see the imbalance data and E$4 is deducted from your balance.
   (b) You see the imbalance data at no cost.
   (c) You do not see the imbalance data and E$4 is deducted.
   (d) You receive E$4 as a bonus.
2. (Treatment only) If you *decline* to purchase data and the round is a **free data round**, what happens?
   (a) You do not see the imbalance data.
   (b) You see the imbalance data and E$4 is deducted from your balance.
   (c) You see the imbalance data at no cost.
   (d) You must pay E$8 to see the data.
3. Is the buy/sell order flow imbalance always useful for predicting the next stock return?
   (a) Yes, it is always informative.
   (b) No, it is never informative.
   (c) It is informative in some rounds and uninformative in others.
   (d) It depends on your investment choice.
4. When do you earn the forecast bonus?

(a) When your forecast is exactly equal to the realized stock return.

(b) When your forecast is within ±1% of the realized stock return.

(c) When your forecast is within ±5% of the realized stock return.

(d) When your forecast is equal to the average stock return of 15%.

5. What happens to the portion of your endowment that you *do not* invest in the stock?

   (a) It earns interest.

   (b) It is lost.

   (c) It remains in cash and stays unchanged.

   (d) It is automatically invested for you.

6. Some participants skim instructions. To show you read this, pick the third option below.

   (a) I skimmed this question.

   (b) I did not read the instructions above.

   (c) I read this and am selecting the third option as instructed.

   (d) I prefer not to answer.

7. How is the portfolio return paid?

   (a) All 12 main days are paid in full.

   (b) One paid day is randomly selected; bonuses accumulate across paid days.

   (c) Only the last day is paid.

   (d) Only training days are paid.

VARIABLES	(1) Return forecast	(2) Return forecast	(3) Investment share
Treated	1.96	1.22	0.13
	(1.15)	(0.86)	(0.98)
Paid round : Belief informative : Imbalance	1.90	2.05**	0.03
	(1.71)	(2.53)	(1.11)
Paid round : Belief uninformative : Imbalance	0.30	-0.07	0.09
	(0.28)	(-0.07)	(1.16)
Belief informative	-0.95	-1.08	
	(-0.60)	(-1.10)	
Belief informative : Imbalance	2.47	2.35	0.07
	(1.74)	(1.53)	(1.33)
Choose to pay	-3.80***	-2.45*	0.03
	(-3.14)	(-2.06)	(0.46)
Choose to pay : Imbalance	1.60**	1.69***	-0.10***
	(2.41)	(3.34)	(-4.28)
Paid round : Belief informative	2.05		0.05*
	(1.61)		(1.84)
Paid round : Belief uninformative	1.84		0.03
	(1.13)		(0.45)
Last return	3.91***	3.70***	0.07***
	(5.47)	(5.70)	(5.91)
Order imbalance	0.18	0.19	0.04*
	(0.16)	(0.17)	(1.84)
Round number	-1.44**	-1.49***	0.02
	(-2.54)	(-3.33)	(1.21)
Overconfidence	1.15		0.17**
	(1.32)		(3.05)
Financial quiz score	1.53		0.20***
	(1.30)		(3.52)
	341		
R-squared	0.29	0.26	0.25

Robust t-statistics in parentheses

*** p<0.01, ** p<0.05, * p<0.1